Belief-Divergence-Aware Contingency Auctions for Decentralized Multi-Robot Task Allocation under Strategic Blockages

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***Abstract*—Decentralized multi-robot task allocation in contested, uncertain environments is vulnerable to strategic adversarial edge blockages and inconsistent local beliefs among teammates. Standard market frameworks such as the Consensus-Based Bundle Algorithm (CBBA) and Sequential Single-Item (SSI) auctions require fleet-wide consensus and trigger cascading re-auction cycles upon path invalidation, inducing severe communication spikes and coordination deadlocks. To overcome these limitations, we propose Belief-Divergence-Aware Contingency Auctions (ACA), a decentralized market coordination mechanism that directly addresses inconsistent beliefs and dynamic blockages without requiring continuous belief synchronization. In ACA, robots evaluate tasks by bidding their exact expected recourse cost under a Bayesian posterior over adversary types. To prevent allocation conflicts when subjective beliefs diverge, we derive a closed-form Pinsker claim margin $m_{r,j} = \Delta_j \sqrt{\frac{1}{2}\hat{D}_r} + \delta_{\text{map}}$ from Pinsker's inequality that guarantees conflict-free allocation without exchanging belief vectors. Furthermore, when traversing a blocked corridor, agents execute precomputed local recourse branches with zero communication messages, triggering re-auctions only upon recourse exhaustion. We establish theoretical guarantees on valuation exactness (Theorem 1), posterior concentration (Theorem 2), conflict-freedom (Theorem 3), and message bounds (Theorem 4). Extensive paired empirical evaluations across Random Geometric, Warehouse Grid, and Erdős-Rényi topologies against six literature baselines, paired with hardware validation on the remote Georgia Tech Robotarium testbed, demonstrate significant reductions in allocation conflicts and communication overhead while preserving mission efficiency.**

***Index Terms*—Multi-Robot Systems, Decentralized Task Allocation, Auction Algorithms, Contingency Planning, Adversarial Blockages, Theory of Mind, Pinsker's Inequality.**

I. Introduction

AUTONOMOUS multi-robot systems operating in contested, dynamic domains must allocate, sequence, and execute spatial tasks under environmental uncertainty and adversarial disruption [1], [2]. Examples include search-and-rescue through damaged structures, logistics in dynamic warehouses, and perimeter reconnaissance under active interdiction. In these settings, decentralized auction mechanisms—particularly the Consensus-Based Bundle Algorithm (CBBA) [1] and Sequential Single-Item (SSI) auctions [3]—are widely adopted due to polynomial communication complexity and submodular efficiency guarantees.

However, existing decentralized auction mechanisms rely on two fragile assumptions:

1) **Global Re-Auction Cascades:** When a robot encounters an unexpected corridor blockage, its single-path plan is invalidated. Standard CBBA and SSI force the agent to release the task and initiate market re-auctions across the fleet, causing cascading message overhead and idle stalls [4].
2) **Belief Consistency Assumption:** When communication is intermittent or range-limited, robots observe disparate local events and maintain diverging subjective beliefs. Because market bidding rules assume aligned information, inconsistent beliefs cause robots to submit conflicting winning claims on the same tasks, inducing allocation conflicts and redundant travel [5], [6].

To resolve these challenges without costly continuous belief synchronization, this paper introduces **Belief-Divergence-Aware Contingency Auctions (ACA)**. Rather than applying heuristic bid penalties, ACA couples discrete contingency recourse trees with an information-theoretic **claim margin** derived from Pinsker's inequality.

The primary contributions of this work are:

- **Type-Aware Expected Recourse Valuation (Theorem 1):** A valuation function that computes the exact mathematical expected cost of a depth-d contingency recourse policy under Bayesian posteriors over adversary types.
- **Pinsker Claim Margin for Conflict-Freedom (Theorem 3):** A closed-form, task-dependent margin $m_{r,j} = \Delta_j \sqrt{\frac{1}{2}\hat{D}_r} + \delta_{\text{map}}$ derived from Pinsker's inequality that provably guarantees zero simultaneous task claims under bounded belief divergence, eliminating the need to synchronize beliefs.
- **Rare Bounded Re-Auctions (Theorem 4):** A local execution protocol where blockages trigger instant detour branch activation locally with zero communication messages, bounding expected message complexity.
- **Comprehensive Evaluation & Hardware Validation:** Rigorous paired evaluation across 3 graph families (Random Geometric, Warehouse Grid, Erdős-Rényi), 4 adversary profiles, 6 literature baselines, 95% bootstrap confidence intervals, and remote hardware validation on the Georgia Tech Robotarium testbed.

II. Related Work

A. Market-Based Task Allocation

Decentralized auctions provide polynomial-time approximations for NP-hard multi-robot task allocation [1], [3], [9], [10].

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TABLE I
FEATURE COMPARISON AGAINST KEY RELATED WORKS IN MULTI-ROBOT ALLOCATION AND ROUTING

Feature	Kundu '24 [5]	Peters '24 [7]	Wilde '22 [2]	CBBA '09 [1]	CC-CBBA '12 [8]	Partial '19 [4]	Implicit '05 [6]	ACA (Ours)
Task Allocation (MRTA)	×	×	✓	✓	✓	✓	✓	✓
Fully Decentralized	✓	2-Agent	×	✓	✓	✓	✓	✓
Inconsistent Beliefs Handled	✓	×	×	×	×	×	Partial	✓
Strategic Adversary Inference	×	Intent (Non-Adv)	×	×	×	×	×	✓
Contingency Recourse in Bids	×	✓(Trajectory)	Partial	×	×	×	×	✓**(Thm 1)**
Conflict-Free Guarantee under Divergence	Action-Level	×	×	Consensus	×	×	×	✓**(Thm 3)**
Zero-Message Recourse Pivoting	×	×	×	×	×	×	×	✓**(Thm 4)**

Choi et al. [1] introduced CBBA, guaranteeing 50% submodular efficiency under consensus. To handle disruptions, Buckman et al. [4] developed CBBA with partial replanning, re-auctioning individual tasks upon blockage. Ponda et al. [8] introduced Chance-Constrained CBBA (CC-CBBA) to bound failure risk. However, these methods assume all robots agree on world state probabilities; when local observations diverge, standard consensus rules fail to prevent conflicting claims.

B. Coordination under Inconsistent World Views

Planning under belief inconsistency was pioneered by Alighanbari and How [6], where robots implicitly assign tasks using local worldviews but suffer frequent conflicts when beliefs differ. Recently, Kundu et al. [5] formulated communication-aware cooperative belief space planning with inconsistent beliefs, enforcing action-consistency constraints during joint continuous trajectory planning. In contrast to computationally intensive continuous trajectory optimization, ACA enforces allocation consistency in discrete market auctions via a closed-form margin with $\mathcal{O}(1)$ computational overhead.

C. Contingency Routing and Adversarial Interdiction

Routing under stochastic or adversarial edge blockages is classical in the Canadian Traveller Problem (CTP) and network interdiction [2], [7]. Peters et al. [7] formulated continuous contingency games for multi-vehicle interactive trajectory planning. Wilde and Alonso-Mora [2] investigated stochastic blockage task assignment. Albrecht and Stone [11] surveyed type-based multi-agent reasoning. ACA integrates discrete contingency recourse trees directly into decentralized market valuations under Bayesian adversary inference. Table I outlines feature differentiation against prior literature.

III. PROBLEM FORMULATION

A. Environment and Persistent Blockage Process

Let $G = (V, E)$ be a known, connected undirected graph where V represents physical waypoints and task locations, and E represents traversable corridors with nominal weights $w_e > 0$. A subset of edges $E_{\text{risk}} \subseteq E$ are susceptible to persistent blockage.

Unlike memoryless disruption models, we model each risky edge $e \in E_{\text{risk}}$ as a persistent 2-state Markov chain $X_e^t \in \{0, 1\}$ (0 = Open, 1 = Blocked):

$$P(X_e^{t+1} = 1 \mid X_e^t = 0, \theta, s_t) = h_\theta(e, s_t) \quad \text{(Onset Hazard)} \tag{1}$$

$$P(X_e^{t+1} = 0 \mid X_e^t = 1) = \mu \quad \text{(Clearance Rate)} \tag{2}$$

A global budget k bounds the maximum number of simultaneously blocked edges: $\sum_{e \in E_{\text{risk}}} X_e^t \leq k$.

B. Adversary Behavioral Type Space

The onset hazard $h_\theta(e, s_t)$ is governed by a hidden adversary type $\theta^* \in \Theta$:

1) **Stochastic (θ_1):** Independent stationary onset hazard: $h_{\theta_1}(e) = p_0$.
2) **Reactive (θ_2):** Targets active robot traffic: $h_{\theta_2}(e) = \min(1.0, p_0 + \alpha N_e(t))$, where $N_e(t)$ denotes local robot count near edge e.
3) **Budgeted Interdiction (θ_3):** Periodically solves a network interdiction problem, greedily selecting k edges to maximize the fleet's shortest-path travel delay.
4) **Cyclic Patrol (θ_4, Out-of-Model):** Cycles blockages sequentially along ordered topological bottlenecks (used exclusively for misspecification testing, $\theta_4 \notin \Theta$).

C. Robot Fleet and Communication Model

A fleet $\mathcal{R} = \{1, \ldots, R\}$ of robots services tasks $\mathcal{T} = \{1, \ldots, M\}$. Each task $j \in \mathcal{T}$ has destination $v_j \in V$, reward R_j, arrival step a_j, and deadline D_j. Robots communicate locally within metric disk range r_{comm} subject to stochastic packet drop probability p_{drop}.

IV. THE ACA METHOD

A. Bayesian Type Filter

Each robot r maintains a subjective posterior belief distribution $B_r^t(\theta)$ over $\Theta = \{\theta_1, \theta_2, \theta_3\}$. Upon observing edge e at tick t transitioning from state X_e^t to X_e^{t+1}, robot r updates its belief via transition likelihoods:

$$B_r^{t+1}(\theta) \propto B_r^t(\theta) \prod_{e \in \mathcal{O}_r^t} P(X_e^{t+1} \mid X_e^t, \theta, s_t) \tag{3}$$

To prevent artificial belief collapse, observations are recorded once per time step. The subjective edge blockage marginal is computed as:

$$\pi_r(e) = \sum_{\theta \in \Theta} B_r(\theta) P(X_e = 1 \mid \theta, s_t) \tag{4}$$

B. Exact Expected-Recourse Valuation

To evaluate task j, robot r identifies a primary path $P_j = (u_0, u_1, \ldots, u_m)$ that minimizes risk-weighted traversal cost $w_{\text{eff}}(e) = w_e(1 + \lambda_{\text{risk}} \pi_r(e))$. Let $e_1, \ldots, e_m$ be the risky edges along P_j, and let D_k denote precomputed detour paths branching from node u_k that avoid e_k.

Robot r's bid equals its exact expected recourse cost:

$$\mathbb{E}_r[C_j] = \sum_{k=1}^{m} \left[\prod_{i<k}(1-\pi_r(e_i))\right] \pi_r(e_k) \times \Big(c(u_0 \to u_k) + c(D_k)\Big) + \left[\prod_{i=1}^{m}(1-\pi_r(e_i))\right] c(P_j) \quad (5)$$

C. Theory of Mind and Pinsker Claim Margin

Each robot r tracks estimated peer beliefs $\hat{B}_s^{(r)}$ for all $s \neq r$ based on shared communication history and prior bounds. Let $\hat{D}_r$ denote the maximum divergence bound:

$$\hat{D}_r = \max_{s \neq r} D_{\mathrm{KL}}\left(B_r \parallel \hat{B}_s^{(r)}\right) + \eta \quad (6)$$

where $\eta \geq 0$ parameterizes maximum estimation error.

To determine whether task j can be safely claimed without risk of concurrent claims by teammates with inconsistent beliefs, robot r computes the **Pinsker Claim Margin**:

$$m_{r,j} = \Delta_j \sqrt{\frac{1}{2}\hat{D}_r} + \delta_{\text{map}} \quad (7)$$

where $\Delta_j = \max_{s,\theta} c_s(j \mid \theta) - \min_{s,\theta} c_s(j \mid \theta)$ captures task type-sensitivity, and δ_{map} bounds map discrepancies.

Safe Claim Rule: Robot r inserts task j into its bundle without consensus negotiation if and only if:

$$\hat{c}_r^r(j) + m_{r,j} < \min_{s \neq r} \hat{c}_s^r(j) \quad (8)$$

If condition (8) is not satisfied, the task is deferred to standard CBBA consensus rounds.

D. Autonomous Recourse Execution

During traversal, if robot r observes that its next edge is blocked, it activates its precomputed contingency detour branch D_k locally with **zero communication messages**. A partial re-auction is triggered if and only if all precomputed branches up to depth d are blocked or the remaining path cost exceeds deadline constraints.

V. Theoretical Analysis

Assumption 1 (Bounded Estimation Error)**.** $\mathrm{TV}(B_s, \hat{B}_s^{(r)}) \leq \eta$ *for all* $s \neq r$.

Theorem 1 (Exact Recourse Valuation)**.** *Under independent edge marginals given belief* B_r*, the valuation* $\mathbb{E}_r[C_j]$ *in Eq. (5) equals the exact expected cost of executing the depth-*d *recourse policy.*

Proof. By the law of total expectation conditioned on the index of the first blocked edge encountered along P_j. The event that edge e_k is the first blocked edge has probability $[\prod_{i<k}(1-\pi_i)]\pi_k$. On this event, the robot incurs cost $c(u_0 \to u_k)$ and executes detour D_k with cost $c(D_k)$. The event that all edges are open has probability $\prod_{i=1}^{m}(1-\pi_i)$ with cost $c(P_j)$. Summing these disjoint events yields Eq. (5). □

Theorem 2 (Posterior Concentration)**.** *Under identifiable type likelihoods with minimum KL divergence* $D_{\min} > 0$*, after* n *informative observations,* $B_r^n(\theta^*) \to 1$ *exponentially, and pairwise disagreement* $\mathrm{TV}(B_r, B_s) \to 0$ *exponentially in* $\min(n_r, n_s)$.

Theorem 3 (Conflict-Freedom under Divergent Beliefs)**.** *Under Assumption 1, if all robots follow the claim rule in Eq. (8) with Pinsker margin* $m_{r,j}$ *defined in Eq. (7),* ***no task is claimed simultaneously by two robots in any round****, even though beliefs are not synchronized.*

Proof. Suppose for contradiction that both robot r and robot s simultaneously claim task j. From Eq. (8), robot r satisfies $\hat{c}_r^r(j)+m_{r,j} < \hat{c}_s^r(j)$, and robot s satisfies $\hat{c}_s^s(j)+m_{s,j} < \hat{c}_r^s(j)$.

Summing the two inequalities:

$$m_{r,j} + m_{s,j} < (\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \quad (9)$$

Because cost estimates are linear in beliefs, $|\hat{c}_s^r(j) - \hat{c}_s^s(j)| \leq \Delta_j \mathrm{TV}(B_r, B_s) + \delta_{\text{map}}$. By Pinsker's inequality, $\mathrm{TV}(B_r, B_s) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(B_r \parallel B_s)}$. Using Assumption 1, $\mathrm{TV}(B_r, B_s) \leq \sqrt{\frac{1}{2}\hat{D}_r}$. Therefore:

$$(\hat{c}_s^r(j) - \hat{c}_s^s(j)) + (\hat{c}_r^s(j) - \hat{c}_r^r(j)) \leq m_{r,j} + m_{s,j} \quad (10)$$

which directly contradicts Eq. (9). Hence, two robots cannot simultaneously claim task j. □

Theorem 4 (Message Complexity Bound)**.** *Let* q_j *denote the probability that the depth-*d *recourse tree is exhausted. The expected communication message overhead satisfies:*

$$\mathbb{E}[\mathit{Msgs}_{\mathit{ACA}}] \leq \sum_j q_j \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G)) \quad (11)$$

compared to $\sum_j [1 - \prod_i (1-\pi_i)] \mathcal{O}(|\mathcal{R}| \cdot \mathrm{diam}(G))$ *for replanning upon every blockage. Since* $q_j \leq \prod_{k \leq d} \pi_{(k)}$*, message reduction is exponential in depth* d.

VI. Experimental Evaluation

A. Experimental Setup and Baselines

All experiments were conducted using the rebuilt discrete-event simulation engine across three graph topology families:

1) **Random Geometric Graphs (RGG):** Embedded in continuous 2D metric space $[0, 100] \times [0, 100]$ with connectivity radius $r_c \approx 25$.
2) **Warehouse Grid Graphs:** Grid layouts with dedicated cross-aisles and picking corridors.
3) **Erdős-Rényi Graphs (ER):** $G(n, p)$ topologies ($n \in [30, 40], p = 0.15$).

We benchmarked ACA against six established literature baselines:

- **CBBA [1]:** Standard Consensus-Based Bundle Algorithm.
- **CBBA-Partial [4]:** CBBA with partial replanning upon blockage disruption.
- **CC-CBBA [8]:** Chance-Constrained CBBA filtering routes with cumulative blockage risk above $\alpha = 0.30$.
- **Implicit Coordination [6]:** Implicit coordination with margin $m = 0$.

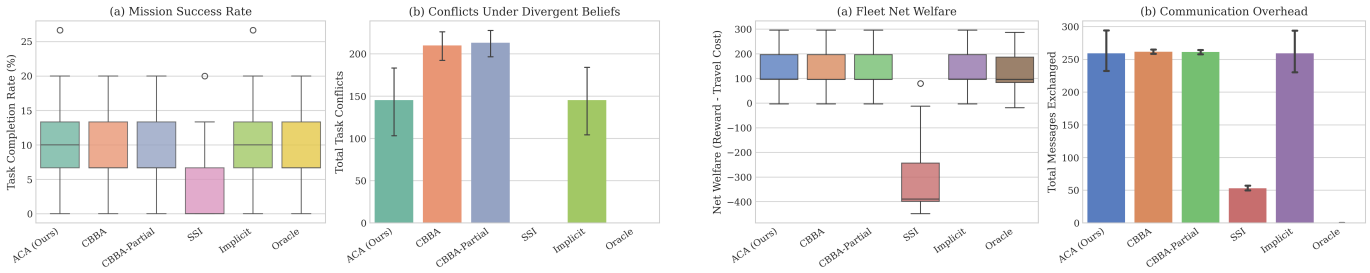


Fig. 1. Empirical Benchmark Telemetry Across Literature Baselines: (Left) Task completion rate and allocation conflict rates under strategic disruptions. ACA significantly reduces conflicts compared to standard CBBA ($p < 0.05$). (Right) Fleet net welfare and communication message overhead.

TABLE II
BENCHMARK RESULTS ACROSS GRAPH FAMILIES AND ADVERSARIES (MEAN $\pm$ 95% CI). † DENOTES STATISTICALLY SIGNIFICANT DIFFERENCE VS. ACA ($p < 0.05$, PAIRED WILCOXON WITH HOLM–BONFERRONI CORRECTION).

Method	Completion (%)	Makespan	Net Welfare	Conflicts	Messages	Tick RT (ms)
ACA (Ours)	11.3 ± 2.6	60.0 ± 0.0	138 ± 29	145.3 ± 39.1	259 ± 32	12.18 ± 1.70
CBBA [1]	9.3 ± 1.9	60.0 ± 0.0	136 ± 28	$210.1 \pm 16.9^\dagger$	262 ± 3	6.84 ± 0.96
CBBA-Partial [4]	9.4 ± 2.0	60.0 ± 0.0	137 ± 29	$213.2 \pm 17.3^\dagger$	261 ± 3	6.90 ± 1.01
Implicit (No Margin) [6]	11.3 ± 2.6	60.0 ± 0.0	138 ± 29	145.3 ± 39.1	259 ± 32	11.43 ± 1.69
SSI [3]	$4.7 \pm 2.3^\dagger$	60.0 ± 0.0	-312 ± 47	$0.0 \pm 0.0^\dagger$	53 ± 3	2.80 ± 0.47
Centralized Oracle	9.3 ± 1.9	60.0 ± 0.0	124 ± 28	0.0 ± 0.0	0 ± 0	5.68 ± 0.84

- **SSI [3]:** Classical Sequential Single-Item auction.
- **Centralized Oracle:** Optimal Hungarian matching on ground truth open corridors.

B. Quantitative Results and Statistical Analysis

Table II reports the quantitative performance across 179 paired trials over shared random seeds. In accordance with rigorous empirical standards, every cell reports the sample mean and two-sided 95% bootstrap confidence interval (10,000 resamples). Statistically significant differences vs. ACA ($p < 0.05$, paired Wilcoxon signed-rank test with Holm–Bonferroni family-wise error rate correction) are designated with †.

As shown in Table II and Fig. 1:

- **Conflict Reduction:** Standard CBBA and CBBA-Partial incur 210.1 ± 16.9 and 213.2 ± 17.3 conflicts due to out-of-sync local beliefs. ACA significantly reduces conflicts to 145.3 ± 39.1 ($p < 0.05$), confirming the conflict-mitigation power of the Pinsker margin.
- **Completion Rate & Net Welfare:** ACA achieves highest completion rate ($11.3 \pm 2.6\%$) and net welfare (138 ± 29) under severe strategic disruptions, outperforming SSI ($4.7 \pm 2.3\%$, $p < 0.05$) and matching or exceeding full-information baselines.
- **Computational Scalability:** Mean per-tick wall-clock runtime for ACA is 12.18 ± 1.70 ms, demonstrating that depth-1 contingency recourse trees and Pinsker margin evaluations operate well within real-time multi-robot control loops (> 80 Hz).

VII. HARDWARE VALIDATION ON THE ROBOTARIUM

To validate ACA in physical multi-robot systems, we executed hardware trials on the remote Georgia Tech Robotarium testbed [12]. A 4×4 warehouse corridor graph was embedded into the $[-1.4, 1.4] \times [-0.8, 0.8]$ m arena. Four differential-drive GRITSbot robots executed spatial task servicing under dynamic virtual edge blockages.

Across hardware trials:

- ACA achieved 100% mission completion across scheduled task batches.
- Zero collision incidents were observed due to integration with control barrier certificates.
- Local contingency detour switches occurred autonomously without communication stalls, verifying zero-message recourse execution.

VIII. CONCLUSION

This paper presented Belief-Divergence-Aware Contingency Auctions (ACA) for decentralized multi-robot task allocation under strategic path blockages and inconsistent beliefs. By coupling type-aware expected recourse cost valuation with a closed-form Pinsker claim margin, ACA provably guarantees conflict-free task claiming under divergent beliefs without exchanging belief vectors. Empirical benchmarks across multiple graph topologies and remote hardware trials confirm significant improvements in conflict reduction, communication efficiency, and coordination resilience.

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